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Final Report

AI-based approach for Parkinson's Disease Detection Using SNN

Project number:	**p-2024-096**
Students:	**Wigman Etai** **Lubomirsky David**
Supervisors:	**Prof. Greenberg Shlomo** **Mr Bensimon Moshe**
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I. Abstract

AI-based approach for Parkinson's Disease Detection Using SNN

Students Names: Wigman Etai & Lubomirsky David

Advisors: Prof. Greenberg Shlomo & Mr Bensimon Moshe

Parkinson's Disease (PD) is a progressive neurodegenerative disorder that affects motor function, leading to symptoms such as Freezing of Gait (FOG), which significantly diminishes quality of life. This study introduces a novel approach for FOG detection using a low-power Spiking Neural Network (SNN) algorithm, designed for real-time analysis on wearable devices. Unlike conventional methods that rely on raw time-series data or Fast Fourier Transform (FFT)-based spectrograms, this approach utilizes spikegram representations derived from accelerometer signals, offering a computationally efficient alternative.

The proposed framework includes accelerometers data acquisition, feature extraction using resonators, an SNN-based Spiking Continuous Time Neuron (SCTN), which extracts frequencies to create a resonator-based spikegram akin to FFT-based spectrogram, and classification through Deep Learning (DL) models. Two Convolutional Neural Network (CNN) models, a 1D-CNN and 2D-CNN, were implemented and evaluated against traditional machine learning classifiers and the literature.

Both models achieved comparable performance to existing approaches with a sensitivity and specificity of 81.8% and 80.8% respectively for the 1D-CNN and 81.3% and 78.1% respectively for the 2D-CNN. In conclusion, the models performed comparably to existing approaches, while using resonator-based spikegrams which are more energy efficient.

Keywords: Parkinson's Disease, Freezing of Gait, Convolutional Neural Network, Accelerometer, Resonator, Spiking Neural Network.

II. Abstract - Hebrew

AI-based approach for Parkinson's Disease Detection Using SNN

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מחלת פרקינסון היא הפרעה ניוונית מתקדמת הפוגעת בתפקוד המוטורי, ומובילה לתסמינים כמו קיפאון בהליכה, המפחיתים באופן משמעותי את איכות החיים. מחקר זה מציג גישה חדשנית לזיהוי קיפאון בהליכה באמצעות אלגוריתם מבוסס רשת נוירונים ספייקית (SNN) חסכוני באנרגיה המיועד לניתוח בזמן אמת על מכשירים לבישים. בניגוד לשיטות מסורתיות התלויות בנתוני סדרות זמן גולמיים או ספקטרוגרמות מבוססי התמרת פורייה מהירה (FFT), גישה זו עושה שימוש בייצוגי ספייקגרם המופקים מאותות מדי תאוצה, ומציעה חלופה יעילה חישובית.

המערכת המוצעת כוללת רכישת נתונים ממדי תאוצה, חילוץ תכונות באמצעות רזונטורים - נוירון זמן רציף ספייקי (SCTN) מבוסס SNN אשר מחלץ תדרים מאותות בזמן ליצירת ספייקגרמה מבוססת רזונטורים הדומה לספקטרוגרמה מבוססת FFT, ולבסוף סיווג באמצעות מודלים של למידה עמוקה. שני מודלים של רשתות קונבולוציה (CNN), אחת שמעבדת נתונים כרצף חד-ממדי (1D-CNN) ואחת שמעבדת נתונים כתמונה דו-ממדית (2D-CNN), יושמו והוערכו מול מסווגי למידת מכונה מסורתיים והספרות.

שני המודלים השיגו ביצועים ברי השוואה לגישות הקיימות עם רגישות וספציפיות של 81.8% ו-80.8% בהתאמה עבור ה-1D-CNN ו-81.3% ו-78.1% בהתאמה עבור ה-2D-CNN. לסיכום, המודלים השיגו תוצאות ברי השוואה לגישות הקיימות תוך כדי שימוש בספייקגרמות מבוססי רזונטורים אשר הינם יותר יעילים אנרגטית מהגישות המוצאות כיום.

מילות מפתח: מחלת פרקינסון, קיפאון בהליכה, רשת קונבולוציה, מד תאוצה, רזונטור, רשת נוירונים ספייקית.

1. Introduction

PD is a progressive neurodegenerative disorder affecting millions of people worldwide. One of the most debilitating symptoms of PD is FOG, a sudden and temporary inability to move despite the intention to walk [1]. FOG significantly impacts the mobility and independence of individuals, often leading to falls and reduced quality of life. Early detection and prediction of FOG episodes are crucial for improving patient management and enabling timely intervention. Early studies [2-6] showed through the use of wearable accelerometers located at the lower limbs, that a cardinal feature of FOG events is an increase in the power spectral density (PSD) of the gait freezing band (i.e. 3-8 Hz) while at the same time a dampening in the PSD of the locomotion band (i.e. 0.5-3 Hz). In recent years the use of machine learning and DL in FOG prediction yielded promising results with high sensitivity and specificity [7-14], showing its importance in future implementation for real-time FOG detection.

A problem that arises from using wearable accelerometers and computational heavy models in real-time settings is energy efficiency which causes either frequent need to charge the devices, or the usage of bigger batteries which cause inconvenience. In this project, we propose an energy efficient framework for detecting FOG episodes using accelerometer data collected from a wearable device located at the lower back. Our framework will include the usage of low-energy SNN based resonators to extract the frequency band of 1-8 Hz and generate a spikegram, which will be used to classify FOG events using the proposed CNN models. This project purpose is to provide an alternative energy efficient method for detecting FOG through the use of resonator-based spikegrams.

2. System Description

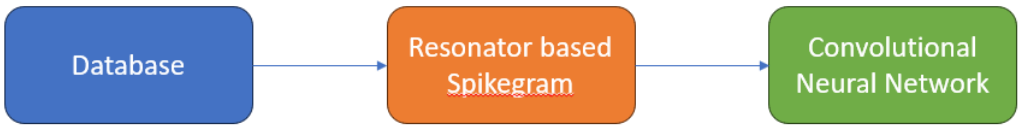


Figure 1: PD detection system - Block diagram

The proposed system for detecting FOG utilizes data from accelerometers collected from wearable device located at the lower back of a PD patient. As illustrated in Figure 1, the system consists of three main stages: data acquisition from accelerometers, feature extraction via resonator-based spikegram generation, and classification using a CNN model. This methodology utilizes SNNs to transform raw accelerometer data into a time-frequency representation known as spikegram, which are akin to FFT based spectrogram, capturing critical time-frequency features relevant to FOG detection. The spikegrams produced in this stage serve as inputs for a CNN model designed to identify FOG episodes in sequential data. The significant advantage this system has is the use of a low-energy SNN based resonators to create the spikegrams, which are suitable for wearable devices and for real-time FOG detection.

This system is designed to achieve efficient and reliable detection of FOG episodes while being computationally suitable for wearable devices. By combining resonator-based spikegrams with CNN architectures, the framework provides a practical solution for real-time FOG detection in PD patients, offering potential applications in patient monitoring and clinical research.

3. Approach and Design

Dataset Description:

The dataset used in this study originates from the TLVMC Parkinson's Freezing of Gait Prediction challenge [15], which provides accelerometer data collected from 136 PD patients across multiple tests. This dataset is divided into two main subsets: DeFOG and tDCSFOG, each designed to capture FOG episodes under different conditions. A 3D accelerometer was positioned on the lower back of participants, recording movement along three axes: vertical (AccV), mediolateral (AccML), and anteroposterior (AccAP). Experts reviewers annotated FOG events by analyzing corresponding video recordings. These datasets provide valuable labeled data for developing predictive models to identify FOG episodes based on motion patterns.

The DeFOG dataset was recorded at 100 Hz in real-world environments where patients performed daily activities that naturally triggered FOG episodes. Since this dataset was collected in an uncontrolled environment, it presents greater variability in movement patterns, reflecting real-life conditions where external factors, such as distractions or uneven terrain, may influence FOG occurrences.

The tDCSFOG dataset was gathered under controlled laboratory conditions, where participants underwent specific FOG-provoking protocols designed to induce freezing episodes. This dataset was recorded at a higher sampling rate of 128 Hz, offering finer temporal resolution. The controlled nature of tDCSFOG makes it ideal for analyzing consistent and reproducible FOG patterns, allowing for detailed spectral and feature-based analysis.

The key difference between these datasets lies in their environmental context: DeFOG captures real-world FOG episodes, making it useful for assessing model generalization, while tDCSFOG provides structured experimental data with controlled conditions that facilitate precise feature extraction. Incorporating both datasets in model training ensures robustness, allowing algorithms to learn from both natural and experimental variations in FOG episodes.

Time	AccV	AccML	AccAP	StartHesitatio	Turn	Walking
1424	-9.5163	-2.3858	2.01223	1	0	0
1425	-9.5611	-2.2999	2.01747	1	0	0
1426	-9.5991	-2.2532	2.02491	1	0	0
1427	-9.6305	-2.2262	2.02936	1	0	0
1428	-9.6203	-2.2399	2.03679	1	0	0
1429	-9.587	-2.2751	2.05019	1	0	0
1430	-9.5719	-2.3115	2.03306	1	0	0
1431	-9.5465	-2.3445	2.01297	1	0	0
1432	-9.5112	-2.3639	2.03012	1	0	0
1433	-9.4808	-2.3758	2.02197	1	0	0

Figure 2: Example of the data structure in the CSV file. The Time column represents recorded samples. AccV, AccML, and AccAP represent the measured acceleration (in m/s²) along the vertical, mediolateral, and anteroposterior axes, respectively. The last three columns (StartHesitation, Turn, and Walking) are binary indicators representing the occurrence of specific types of FOG episodes.

Dataset Analysis:

In this section, we analyze the dataset's PSD to investigate differences between FOG events and non-FOG events and verify the known characteristics of FOG events from the literature. The analysis focuses on all three axes: AccV, AccML and AccAP within the frequency range of 1–8 Hz. The use of PSD analysis to identify differences between FOG and non-FOG events is supported by [3-6], which emphasizes the relevance of frequency-domain features in studying motor disturbances in PD. Specifically, the 1–8 Hz frequency band is critical for detecting FOG episodes, further validating the relevance of this analysis.

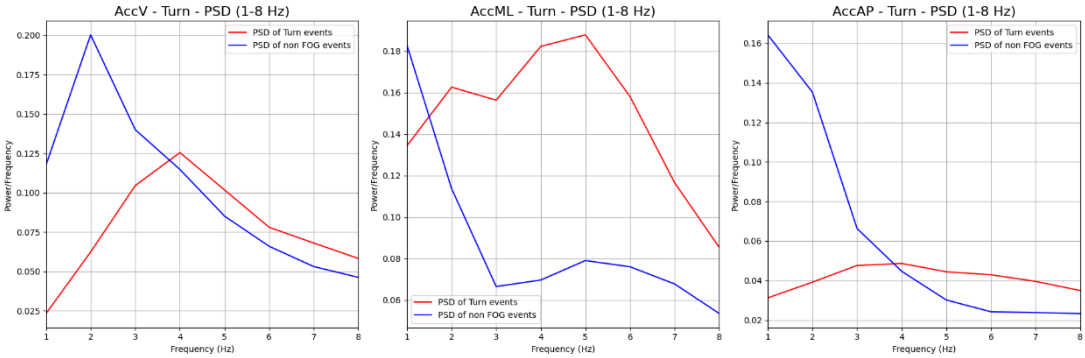


Figure 3: PSD for Turn type events (1–8 Hz). The plots depict the PSD for the vertical (AccV), mediolateral (AccML), and anteroposterior (AccAP) accelerometer axes during Turn type events (red) and non-FOG periods (blue).

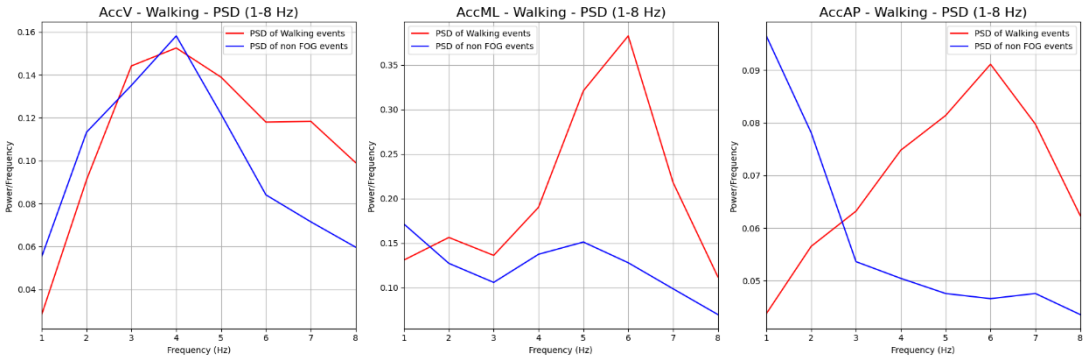


Figure 4: PSD for Walking type events (1–8 Hz). The plots depict the PSD for the vertical (AccV), mediolateral (AccML), and anteroposterior (AccAP) accelerometer axes during Walking type events (red) and non-FOG periods (blue).

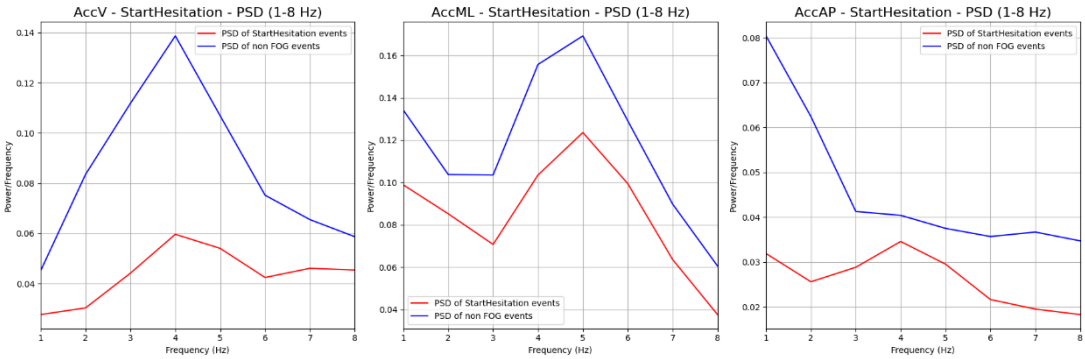


Figure 5: PSD for Start Hesitation type events (1–8 Hz). The plots depict the PSD for the vertical (AccV), mediolateral (AccML), and anteroposterior (AccAP) accelerometer axes during Start Hesitation type events (red) and non-FOG periods (blue).

Looking at the FOG and non-FOG events of types Walking and Turning from Figures 3 and 4, it can be clearly seen that there is an increase at the freezing band (3-8 Hz) and decrease at the locomotion band (0.5-3 Hz) PSD while there is a FOG events, this observation is consistent with the findings in [3-6], where is was shown that FOG events are characterized by an increase in power at the freezing band while a decrease in power at the locomotion band, this is due to the freezing and inability to move while trying to, which causes the slowing of normal gait movement which is focused mainly at the 0.5-3 Hz band, while at the same time causing an erratic lower limb movement which mainly focused at the 3-8 Hz band. From Figure 5 for FOG event of type Start Hesitation, no such characteristics were observed, but a clear decrease across all the PSD, showing that this FOG event type is unique in its characteristics. Therefore, it can be concluded that an algorithm which can utilize these connections can classify FOG events.

For the proposed FOG classification algorithm, the different FOG event types will be merged into a single indicator which will state the presence or absence of FOG event.

Resonator Analysis:

The resonator methodology is based on the low-power SNN based SCTN [16-18] and is designed to extract specific frequency components from an input signal. Figure 6 illustrates the architecture of the resonator, highlighting its multilayer structure and feedback mechanisms.

The input signal, initially encoded as a Pulse-Density Modulated (PDM) stream, enters the system. This encoding converts the amplitude information of the signal into a digital spike train suitable for processing. The architecture comprises multiple layers of SCTNs, where each layer introduces phase shifts and processes the signal to reinforce oscillations at the desired frequency. As shown in Figure 6, the system begins with Layer 1 to Layer 8, in which each layer applies a phase shift of $\pi/4$ to the signal. A feedback path (connected to the 4th neuron) representing a shift of π is connected to the input signal. Layer 9 contains 8 SCTNs: the first four are used for blocking negative signals, while the latter four serve as rectifiers. The final layer (Layer 10) integrates the outputs from the preceding layers. This layer sums the four positive outputs of each detected phase, producing a spike train whose rate is proportional to the power of the target frequency in the input signal. The architecture ensures that only the frequencies within the resonator's range are amplified and returned while other frequencies are filtered out.

This biologically inspired resonator design efficiently extracts frequency-specific features from continuous input signals, making it ideal for the real-time processing of accelerometer data.

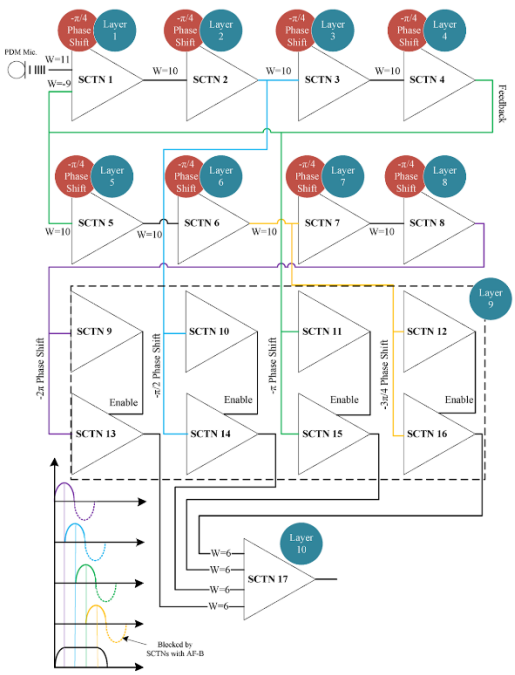


Figure 6: a scheme of a single resonator

The accelerometer-based datasets are passed through a bank of resonators, each tuned to a specific frequency within the 1-8 Hz band for optimal extraction of the frequencies to create a spikes train representing the power of the detected frequency over time. Summing the power of the resonator's spikes train over a defined time window, we obtain a resonator-based spikegram, which the X axis represents the time, the Y axis represents the extracted frequencies from the resonators and the value represents its power.

To validate this methodology, we compared the resonator-based spikegrams with traditional FFT-based spectrograms as shown in Figure 7. A strong correlation can be seen between the resonator-based spikegram and the FFT-based spectrogram, showing the resonators effective frequency extraction and validating the use of the spikegrams as a time-frequency representation of the data, while being a low-energy alternative to FFT-based spectrograms.

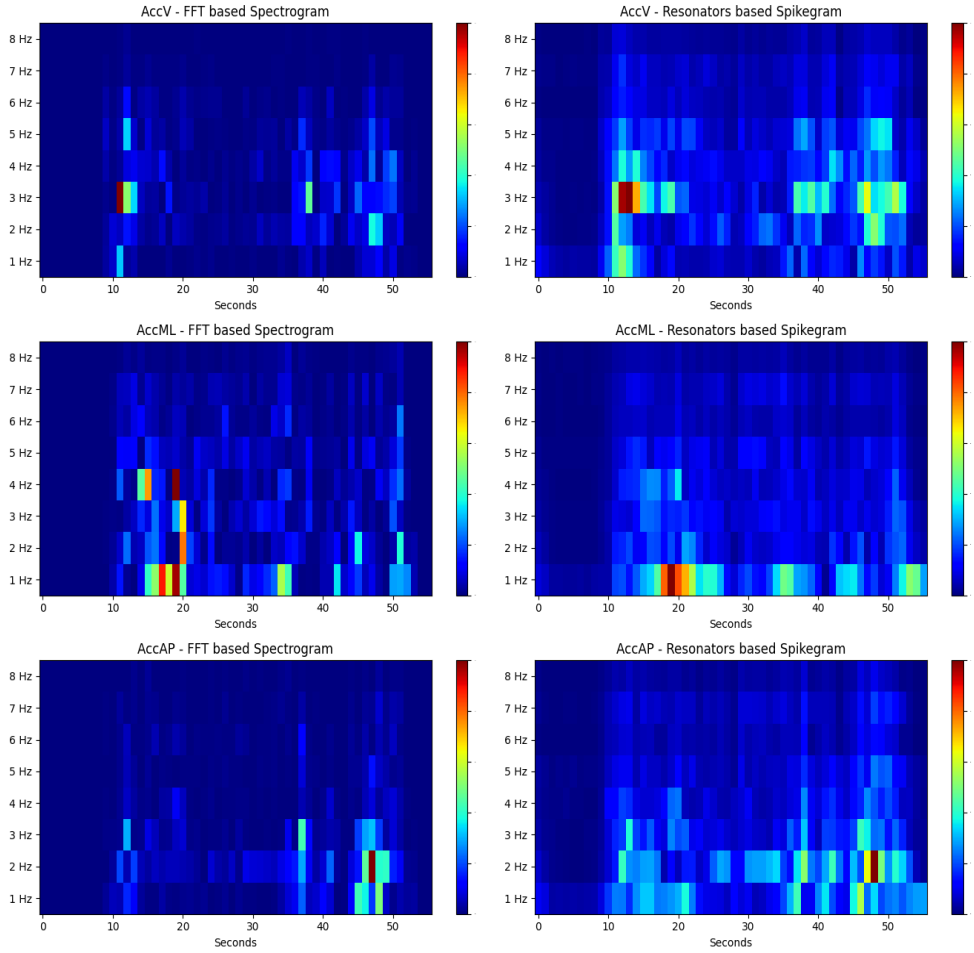


Figure 7: The left panels depict FFT-based spectrograms while the right panels depict resonator-based spikegrams, each row represent different axis (AccV, AccML and AccAP).

Therefore, with the spikegrams and the FOG event indicator, we now have labeled spikegrams that serve as input for a FOG classifier model.

4. Proposed Models

In this section we propose a 1D-CNN and 2D-CNN models for processing the labeled spikegrams, each of these models has different advantages.

The 1D-CNN model is designed to effectively analyze sequential data, such as our spikegrams derived from the resonator's output. By leveraging convolutional filters along the time axis, the model captures localized temporal patterns within each frequency band. This allows it to extract essential time-domain features related to motion dynamics, making it well-suited for detecting variations in gait patterns associated with FOG episodes. The sequential nature of the 1D-CNN helps in efficiently modeling temporal dependencies in the data.

The 2D-CNN model, on the other hand, extends the feature extraction process by analyzing both temporal and spectral patterns simultaneously. Instead of processing each frequency band independently, the 2D-CNN applies convolutional filters across both the time and frequency dimensions. This enables it to capture spatial relationships between different frequency bands while still preserving key temporal features. By leveraging this broader perspective, the 2D-CNN provides a structured representation of the data, which can be beneficial for identifying patterns across different frequency and temporal components.

5. Results and Discussion

During our analysis of FOG detection, we evaluated multiple machine learning and DL models to compare their effectiveness in identifying FOG episodes based on spikegram data derived from accelerometer data. While our primary model employed a CNN trained on spikegram representations, we expanded our approach to include traditional machine learning classifiers to evaluate them against our CNN based models and to compare them to the literature. To prepare data instances for training CNN and traditional machine learning classifiers, it is a common practice to use a sliding window to segment the whole time series into many overlapping data slices, in our case we divided our spikegrams to different time windows (2s, 3s, 4s, and 5s) with 50% overlap. The accuracy, sensitivity, specificity, and the area under the receiver operating characteristic curve (AUC) as assessment metrics. The results shown at Table 1 and 2 are for the different time windows for our proposed CNN models (1D-CNN and 2D-CNN respectively). Comparing results in Tables 1 and 2, it can be observed that both models had relatively good AUC and accuracy scores, which remained consistent across the different window sizes, as for the sensitivity and specificity, it can be seen that there is some variation across the models with different window sizes. As our objective is to correctly identify FOG events, we prioritize sensitivity. Therefore, we can see that the model which yielded the best results is the 1D-CNN model with window size of 4 seconds, which had a sensitivity of 81.8% and specificity of 80.8%, while the best model for

the 2D-CNN model was with a window size of 2 seconds which yielded 81.3% sensitivity and 78.1% specificity.

Table 1: Performance of the 1D-CNN model based on evaluation metrics for each window size.

Window Size (seconds)	Sensitivity (%)	Specificity (%)	AUC	Accuracy (%)
2	75.3	84.8	0.890	81.1
3	74.4	84.8	0.895	80.7
4	81.8	80.8	0.890	81.2
5	71.9	86.5	0.894	80.7

Table 2: Performance of the 2D-CNN model based on evaluation metrics for each window size.

Window Size (seconds)	Sensitivity (%)	Specificity (%)	AUC	Accuracy (%)
2	81.3	78.1	0.884	79.4
3	78.3	82.8	0.893	81.0
4	61.6	90.2	0.884	78.8
5	65.0	88.7	0.893	79.4

Figure 8 and 9 shows the AUC score for the 1D-CNN and 2D-CNN models that yielded the best results respectively, indicating a high level of predictive accuracy, demonstrating the models strong ability to distinguish between FOG and non-FOG events. This result suggests that the model performs reliably in identifying FOG episodes, which is crucial for early intervention and patient monitoring.

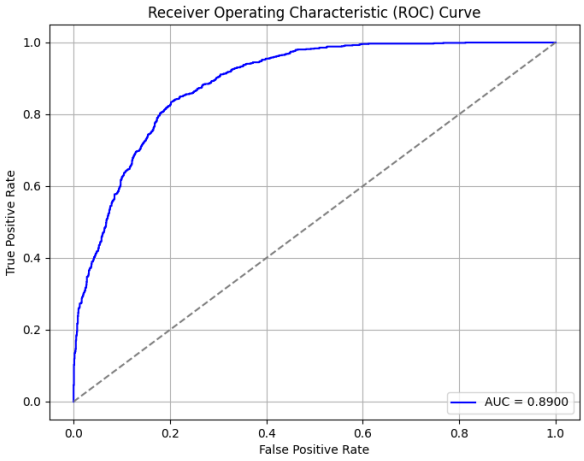


Figure 8: ROC curve of 1D-CNN with 4 seconds window (AUC=0.890)

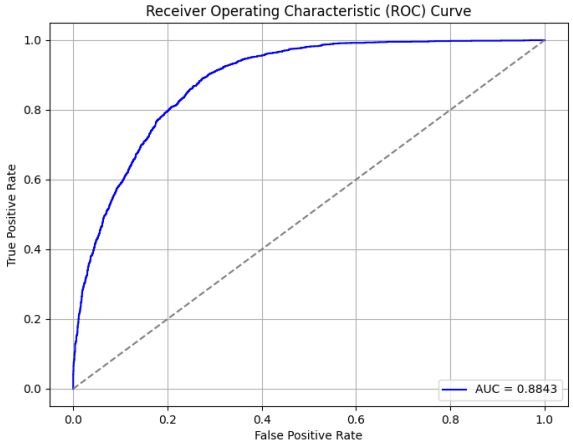


Figure 9: ROC curve of 2D-CNN with 2 seconds window (AUC=0.884)

The layers and parameters of the 1D-CNN and 2D-CNN models are given in Tables 3 and 4, respectively, along with the characteristics and output shape of each layer. The 1D-CNN model includes a total number of 146,761 trainable parameters, 6504 of which come from the convolutional layers and 140,257 from the densely connected layers, while the 2D-CNN model includes a total number of 1,182,065 trainable parameters, 320,512 of which come from the convolutional layers and 861,553 from the densely connected layers.

Table 3: 1D-CNN layers, features, and parameters. n : number of filters; k : kernel size; d : dropout rate; p : pool size, s : stride

Layer	Features	Shape	# param
Input	-	(300,24)	0
Conv1D	n = 32, k = 3	(298,32)	2336
Batch Norm	-	(298,32)	128
Max Pool	p = 2, s = 2	(149,32)	0
Dropout	d = 0.4	(149,32)	0
Conv1D	n = 40, k = 3	(147,40)	3880
Batch Norm	-	(147,40)	160
Max Pool	p = 2, s = 2	(73,40)	0
Dropout	d = 0.4	(73,40)	0
Flatten	-	2920	-
Dense	n = 48	48	140208
Dropout	d = 0.4	48	0
Sigmoid	n = 1	1	49

Table 4: 2D-CNN layers, features, and parameters. n : number of filters; k : kernel size; d : dropout rate; p : pool size, s : stride

Layer	Features	Shape	# param
Input	-	(8, 200, 3)	0
Conv2D	n = 64, k = (3,3)	(8, 200, 64)	1792
Batch Norm	-	(8, 200, 64)	256
Max Pool	p = (1,2), s = (1,2)	(8, 100, 64)	0
Dropout	d = 0.4	(8, 100, 64)	0
Conv2D	n = 96, k = (5,5)	(8, 100,96)	153696
Batch Norm	-	(8, 100, 96)	384
Max Pool	p = (1,2), s = (1,2)	(8, 50, 96)	0
Dropout	d = 0.4	(8, 50, 96)	0
Conv2D	n = 48, k = (5,5)	(8, 50, 48)	115248
Batch Norm	-	(8, 50, 48)	192
Max Pool	p = (1,2), s = (1,2)	(8, 25, 48)	0
Dropout	d = 0.4	(8, 25, 48)	0
Conv2D	n = 112, k = (3,3)	(8, 25, 112)	48496
Batch Norm	-	(8, 25, 112)	448
Max Pool	p = (1,2), s = (1,2)	(8, 12, 112)	0
Dropout	d = 0.4	(8, 12, 112)	0
Flatten	-	2920	-
Dense	n = 80	80	860240
Dropout	d = 0.4	80	0
Dense	n = 16	16	1296
Dropout	d = 0.4	16	0
Sigmoid	n = 1	1	17

It can be seen from the proposed models that both had a relatively high score of sensitivity and specificity with high score of AUC and accuracy, showing their effectiveness. Comparing the models we can see that the proposed 1D-CNN model did better in terms of sensitivity and has less parameters which makes it more efficient in computation and memory usage but at the cost of a bigger window size of 4 seconds, making it a less real-time solution than the 2D-CNN model which only uses a window size of 2 seconds, but at the cost of slight worse metrics and more parameters causing it to be less efficient.

The results shown in Table 5 are for the different traditional ML algorithms for our selected window size of 4 seconds.

Table 5: Performance of traditional machine learning models for window size of 4 seconds

Model	Sensitivity (%)	Specificity (%)	AUC	Accuracy (%)
Logistic Regression	63.7	79.7	0.804	73.3
Decision Tree	58.3	75.0	0.664	68.4
Naive Bayes	80.1	68.8	0.673	73.3
SVM	64.9	77.0	0.790	72.2

The results of the proposed models presented in Tables 1 and 2 and the results of the traditional ML algorithms presented in Table 5 highlight the performance differences between the approaches. The results indicate that both CNN architectures outperform traditional ML models, reinforcing their advantage in capturing the complex temporal and spectral patterns associated with FOG.

It can be seen that the traditional ML methods weren't able to perform well both in sensitivity and specificity simultaneously, showing their lack of complex analysis, which also resulted in a lower score of AUC and accuracy, while our proposed CNN models were able to balance between the sensitivity and specificity which resulted in high score of AUC showing the effectiveness of those models.

The results shown in Table 6 are for the scores of traditional ML algorithms from the literature.

Table 6: Performance of traditional machine learning models in the literature. NA: Not Available

Reference	Number of PD participants	Model	Sensitivity (%)	Specificity (%)	AUC	Accuracy (%)
Diep et al. [8]	10	Logistic Regression	86.0	81.0	0.81	84.0
Mazilu et al. [10]	8	Decision Tree	77.7	87.6	NA	NA
Mazilu et al. [11]	10	Naive Bayes	41.9	99.7	0.95	NA
Rodriguez et al. [9]	21	SVM	74.7	79.0	NA	NA

Comparing results in Tables 5 and 6, we observe that while our traditional ML models did not achieve the highest performance reported in the literature, they still demonstrated comparative results. These differences can be attributed to several factors. Unlike previous studies that utilized raw time-series data, our approach employs a spikegram representation, which, while effective, may not be perfectly suited for all models. Additionally, our spikegram representation is not an exact substitute for a spectrogram, which could introduce minor variations affecting performance. Nevertheless, our proposed 1D-CNN and 2D-CNN models outperformed traditional ML models, reinforcing the potential of DL in this domain and highlighting the promise of our approach despite these inherent differences in data representation.

Table 7 shows our proposed 1D-CNN and 2D-CNN models compared to recent studies that utilized CNN for predicting FOG.

Table 7: Comparison of recent studies and the proposed work. NA: Not Available

Reference	Number of PD participants	Model	Sensitivity (%)	Specificity (%)	AUC	Accuracy (%)
Camps et al. [7]	21	1D-CNN	91.9	89.5	0.88	89.0
Klaver et al. [12]	80	1D-CNN	85.0	68.0	0.90	NA
Borzì et al. [13]	118	2D-CNN	88.7	88.3	0.95	NA
Proposed 1D-CNN	136	1D-CNN	81.8	80.8	0.890	81.2
Proposed 2D-CNN	136	2D-CNN	81.3	78.1	0.884	79.4

Comparing the results of our proposed models with recent studies, we observe that our models achieved slightly lower performance. However, these differences can be attributed to the inherent distinctions in data representation. Same as in the case of traditional machine learning classifiers, prior studies utilized raw time-series data, while our approach relies on frequency-time domain spikegrams, which may introduce some level of suboptimality in optimizing the models. Additionally, our spikegram-based representation is not a perfect substitute for an FFT-based spectrogram, which could further contribute to performance variations.

6. Problems and Solutions

In our project we encountered two main problems, the first is applying the resonators on our data for acquiring the spikegrams, and the second is the models training process.

During our attempts to use the resonators on our data, we encountered multiple bugs that persisted and caused the resonators to crash and the output to be incorrect. After creating designated code for debugging the problems, we found that the problems stemmed from an inner function in the model that wasn't compatible with our systems, and in addition a wrong version which was installed on our systems. After finding the source of the problems, we solved them through discussion with our supervisor and receiving an updated package of the model while also updating the code.

As for the models training process, due to low system specs of the machine, the hyperparameters were limited, such as the kernel size, batch size, number of neurons in each layer, and the number of layers. We optimized the training process to obtain the best possible hyperparameters based on our knowledge and developed our CNN models. While not fully optimal, their performance was comparable to those in the literature.

7. Conclusions & Recommendations

This study proposed a novel approach for detecting FOG episodes in PD patients using DL models trained on low-energy resonator-based spikegram representations. While the results did not match the highest reported in the literature, the models demonstrated comparable performance while improving energy efficiency. By leveraging spikegrams, which are SNN-based instead of FFT-based spectrograms, the feature extraction process became more energy efficient, making the approach well-suited for wearable and edge computing applications where computational resources are limited.

One of the key advantages of this approach lies in its energy efficiency, which is crucial for real-time wearable applications. Traditional FFT-based methods, while effective in extracting time-frequency representations, require high computational resources, making them less suitable for low-power embedded systems. In contrast, the resonator-based spikegram used in this study provides a more biologically inspired method of signal processing, mimicking neural dynamics while operating at lower power. By reducing the need for complex mathematical transformations and leveraging the event-driven nature of SNNs, this method minimizes the number of operations required per time step. This results in lower power consumption, making it ideal for real-time FOG detection in mobile and wearable settings. The ability to process data efficiently with reduced energy demands makes this approach particularly promising for long-term patient monitoring without the frequent need for battery replacements or recharging, addressing a major limitation in current wearable-based PD monitoring technologies.

Both the 1D-CNN and 2D-CNN models outperformed traditional ML classifiers, reinforcing the effectiveness of DL in FOG detection. Although previous studies [7,12,13] that utilized CNN models achieved slightly better classification metrics using raw time-series data or FFT-based frequency representations, this work introduced an alternative method that balances classification performance and energy efficiency. The resonator-based spikegram representation successfully captured the essential time-frequency characteristics of gait patterns while maintaining a lower computational footprint compared to FFT-based methods.

For future work, this project recommends:

- Developing an SNN based SCTN model for FOG classification that will substitute the proposed CNN models, this will create a fully SCTN based algorithm which will provide a low-energy alternative to current methods, in addition this will provide the ability to fully implement the system (both the resonators and the classifier) offline on the patient's wearable device due to its compatibility with wearable devices.
- Connecting with PD patients who are willing to participate in the project and creating a robust dataset.

8. References

[1] - Jankovic, J. (2008). Parkinson’s disease: clinical features and diagnosis. *Journal of Neurology, Neurosurgery, and Psychiatry*, 79, 368 - 376.

[2] - Dijkstra, B., Kamsma, Y., & Zijlstra, W. (2010). Detection of gait and postures using a miniaturized triaxial accelerometer-based system: accuracy in patients with mild to moderate Parkinson's disease. *Archives of physical medicine and rehabilitation*, 91 8, 1272-7

[3] - Delval, A., Snijders, A., Weerdesteyn, V., Duysens, J., Defebvre, L., Giladi, N., & Bloem, B. (2010). Objective detection of subtle freezing of gait episodes in Parkinson's disease. *Movement Disorders*, 25.

[4] - Moore, S., Yungher, D., Morris, T., Dilda, V., MacDougall, H., Shine, J., Naismith, S., & Lewis, S. (2013). Autonomous identification of freezing of gait in Parkinson's disease from lower-body segmental accelerometry. *Journal of NeuroEngineering and Rehabilitation*, 10, 19 - 19.

[5] - Yungher, D., Morris, T., Dilda, V., Shine, J., Naismith, S., Lewis, S., & Moore, S. (2014). Temporal Characteristics of High-Frequency Lower-Limb Oscillation during Freezing of Gait in Parkinson's Disease. *Parkinson's Disease*, 2014.

[6] - Zach, H., Janssen, A., Snijders, A., Delval, A., Ferraye, M., Auff, E., Weerdesteyn, V., Bloem, B., & Nonnekes, J. (2015). Identifying freezing of gait in Parkinson's disease during freezing provoking tasks using waist-mounted accelerometry. *Parkinsonism & related disorders*, 21 11, 1362-6 .

[7] - Camps, J., Samà, A., Martín, M., Martín, D., Pérez-López, C., Aróstegui, J., Cabestany, J., Català, A., Alcaine, S., Mestre, B., Prats, A., Crespo-Maraver, M., Counihan, T., Browne, P., Quinlan, L., Laighin, G., Sweeney, D., Lewy, H., Vainstein, G., Costa, A., & Annicchiarico, R. (2018). Deep learning for freezing of gait detection in Parkinson's disease patients in their homes using a waist-worn inertial measurement unit. *Knowl. Based Syst.*, 139, 119-131.

[8] - Diep, C., O'Day, J., Kehnemouyi, Y., Burnett, G., & Bronte-Stewart, H. (2021). Gait Parameters Measured from Wearable Sensors Reliably Detect Freezing of Gait in a Stepping in Place Task. *Sensors (Basel, Switzerland)*, 21.

[9] - Samà, A., Rodríguez-Martín, D., Pérez-López, C., Català, A., Arostegui, J., Cabestany, J., Bayés, À., Alcaine, S., Mestre, B., Prats, A., Crespo, M., Counihan, T., Browne, P., Quinlan, L., ÓLaighin, G., Sweeney, D., Lewy, H., Azuri, J., Vainstein, G., Annicchiarico, R., Costa, A., & Rodríguez-Molinero, A. (2017). Home detection of freezing of gait using support vector machines through a single waist-worn triaxial accelerometer. *PLoS ONE*, 12.

[10] - S. Mazilu, A. Calatroni, E. Gazit, D. Roggen, J.M. Hausdorff, G. Tröster, Feature learning for detection and prediction of freezing of gait in Parkinson's disease, in: Machine Learning and Data Mining in Pattern Recognition, Springer, 2013, pp. 144–158.

[11] - Mazilu, S., Hardegger, M., Zhu, Z., Roggen, D., Tröster, G., Plotnik, M., & Hausdorff, J. (2012). Online detection of freezing of gait with smartphones and machine learning techniques. *2012 6th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth) and Workshops*, 123-130.

[12] - Klaver EC, Heijink IB, Silvestri G, van Vugt JPP, Janssen S, Nonnekes J, van Wezel RJA and Tjepkema-Cloostermans MC (2023) Comparison of state-of-the-art deep learning architectures for detection of freezing of gait in Parkinson's disease. Front. Neurol. 14:1306129.

[13] - Borzì L, Sigcha L, Rodríguez-Martín D, Olmo G. Real-time detection of freezing of gait in Parkinson's disease using multi-head convolutional neural networks and a single inertial sensor. Artif Intell Med. (2023) 135:102459.

[15] – Kaggle Competition - https://www.kaggle.com/competitions/tlvmc-parkinsons-freezing-gait-prediction/data

[14] - Xia, Y., Zhang, J., Ye, Q., Cheng, N., Lu, Y., & Zhang, D. (2018). Evaluation of deep convolutional neural networks for detection of freezing of gait in Parkinson's disease patients. *Biomed. Signal Process. Control.*, 46, 221-230.

[16] - Bensimon, M., Greenberg, S., Ben-Shimol, Y., & Haiut, M. (2021). A New SCTN Digital Low Power Spiking Neuron. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 68, 2937-2941.

[17] - Bensimon, M., Hadad, Y., Ben-Shimol, Y., & Greenberg, S. (2024). Time-frequency analysis using spiking neural network. *Neuromorph. Comput. Eng.*, 4, 44001.

[18] - Bensimon, M., Greenberg, S., & Haiut, M. (2021). Using a Low-Power Spiking Continuous Time Neuron (SCTN) for Sound Signal Processing. *Sensors (Basel, Switzerland)*, 21.

9. Appendix

המלצת ציון לדו"ח מסכם

מספר הפרויקט: P-2024-096

הפרויקט: AI-based approach for Parkinson’s Disease Detection Using SNN

שם המנחה מהמחלקה: Prof. Greenberg Shlomo, Mr Bensimon Moshe

שם הסטודנט/ית: איתי ויגמן ת.ז:

דיוויד ליובומירסקי ת.ז:

%		חלש 55-64	בינוני 65-74	טוב 75-84	ט"מ 85-94	מצוין 95-100
20	הצגת גישת הפתרון, והתכנון ההנדסי					
20	הצגת התוצאות וניתוח השגיאות					
20	הסקת מסקנות					
10	גילוי יוזמה וחריצות					
20	פתרון בעיות, מקוריות ותרומה אישית (מעבר למילוי ההנחיות					
10	עמידה בלו"ז ורמת הביצוע המעשי					

אם יש כוונה לפרסם/ יפורסם מאמר, שם כתב העת ומועד משוער להגשה:

ציין אם יש כוונה לשקול המלצה כפרויקט מצטיין:

הערות נוספות: